Deep Ensembles

Deep Ensembles[52]

Deep Ensembles approximate uncertainty by training multiple independent networks and averaging their predictions.

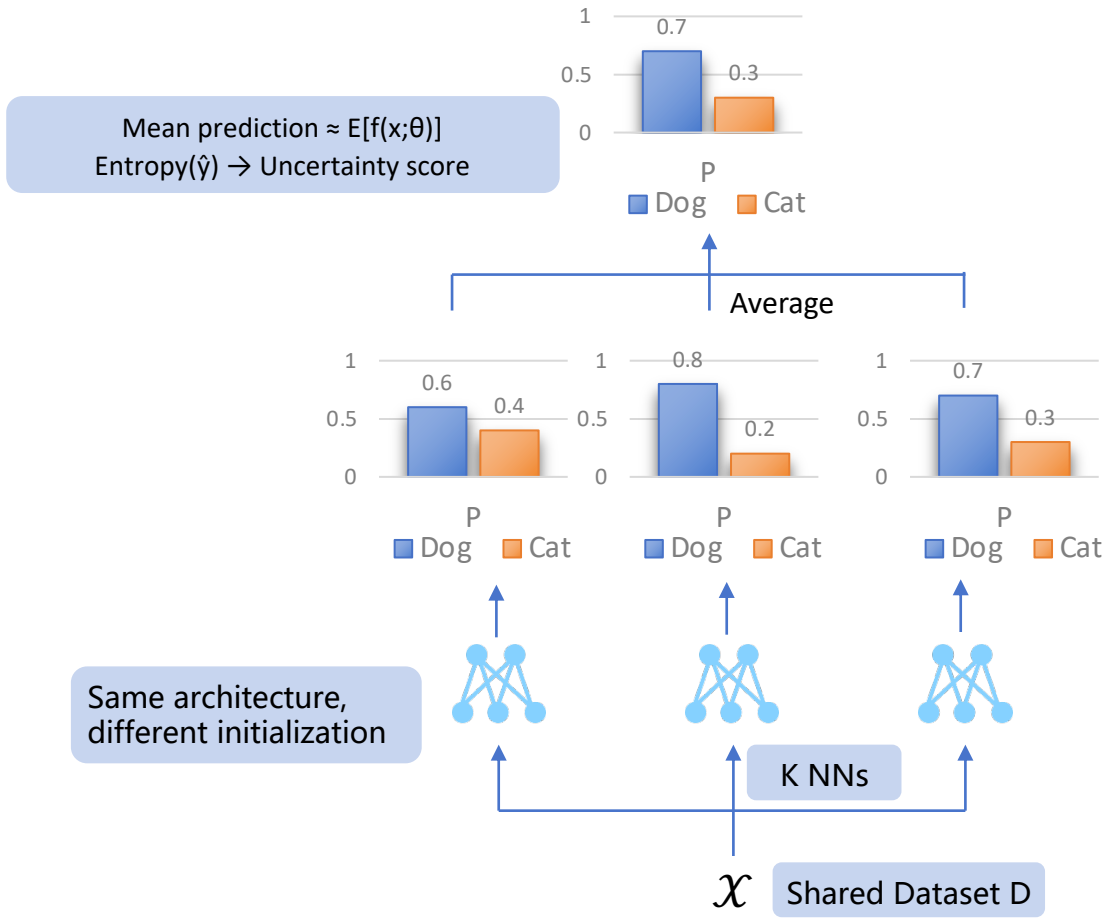
- **Training and Inference Pipeline**
  - Initialize K independent models with different random seeds.
  - Train them in parallel on the same dataset: $\{\theta_1, \theta_2, .., .. \theta_K\}$.
  - For an input x:

$$y_i = f(x; \theta_i), i \in [K]$$

  - Compute ensemble mean and entropy:

$$\hat{y} = \frac{1}{K}\sum_i y_i \,, \hat{H} = \text{Entropy}(\hat{y})$$

-> Higher entropy = higher uncertainty.



[52] Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles, Neurips 2016

Deep Ensembles[52]

- **Task Type:** Regression / Classification task

- **Introduction:** Deep Ensembles approximate uncertainty by training multiple independent networks with different initializations and averaging their predictions. It quantifies parameter uncertainty and establishes a reliable non-Bayesian baseline in modern UQ.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Ensemble	Parameter	White-box	Full training	High	Calibration/Ranking

[52] Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles, Neurips 2016